Prospective low-prevalence evaluation

5,000 new URLs • 2.00% phishing • frozen models and thresholds

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
Weighted HYB-MS real-only	98.38%	57.04%	77.00%	65.53%	96.16%	73.16%	1.18%	23.00%
Weighted HYB-MS +10% GAN	98.64%	62.70%	79.00%	69.91%	96.42%	76.94%	0.96%	21.00%

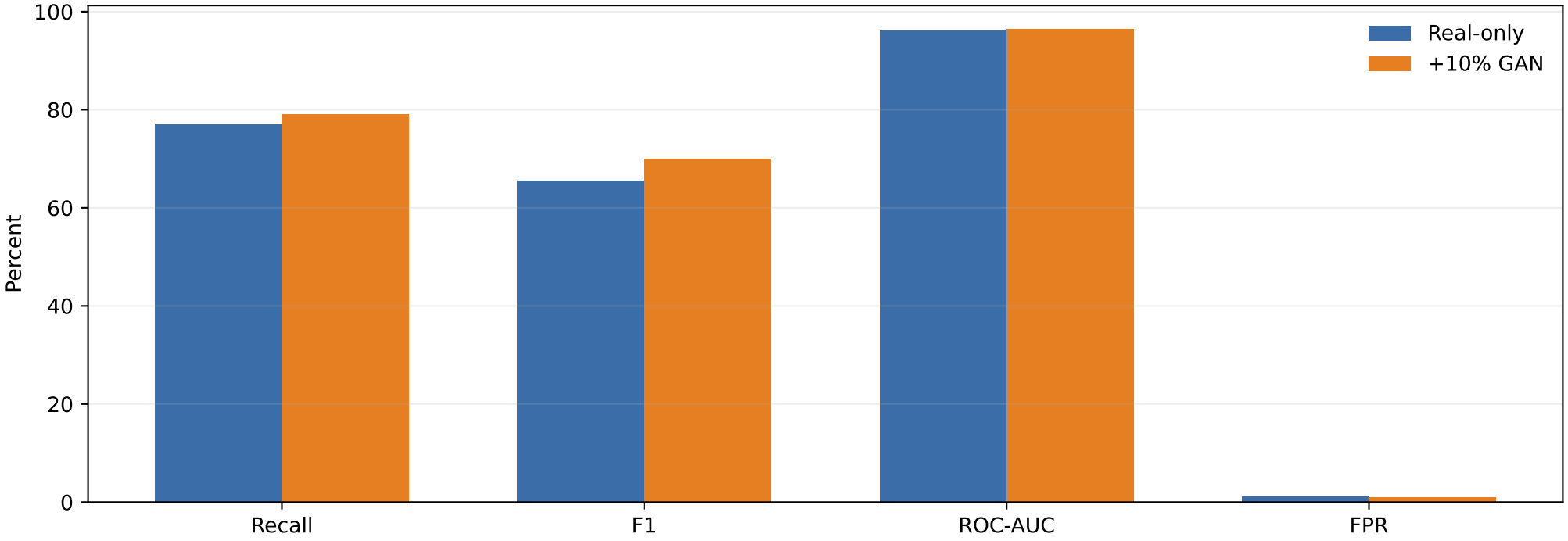
AUGMENTATION BENEFIT SUPPORTED

The +10% model improved recall, F1 and ROC-AUC without increasing FPR on this new cohort.

No training, calibration, checkpoint choice or threshold was changed after this dataset was frozen.

Dataset SHA-256: efa7b15e3b41727002bfb0a36c9de9685495424769b14081b2063156dffffc9c

What the result means in browser-like traffic



Real-only false alerts per 10,000 visits: 116.0

+10% GAN false alerts per 10,000 visits: 94.0

This is an identity-disjoint, proxy-labelled evaluation. It is useful evidence, but it is still not a live production deployment test.